

Machine Learning techniques and models for anomaly detection and prediction in DAQ systems: status and plans

Joaquin Hoya,
with inputs from G. Avolio, S. Chapeland and P. Cifra

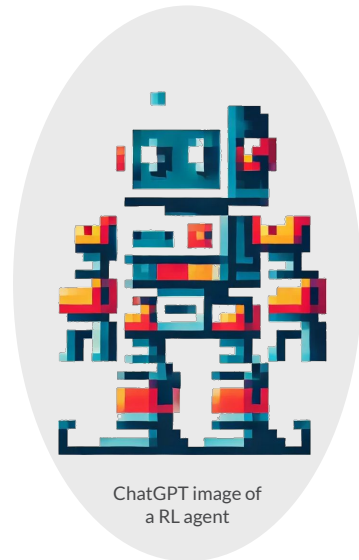


Château de Bossey

04/02/2025

Outline

- Introduction and motivation
- Textbook anomalies in Time Series
- ML methods: Clustering, Deep Learning and evaluation metrics
- Status of the ongoing work
- Future plans

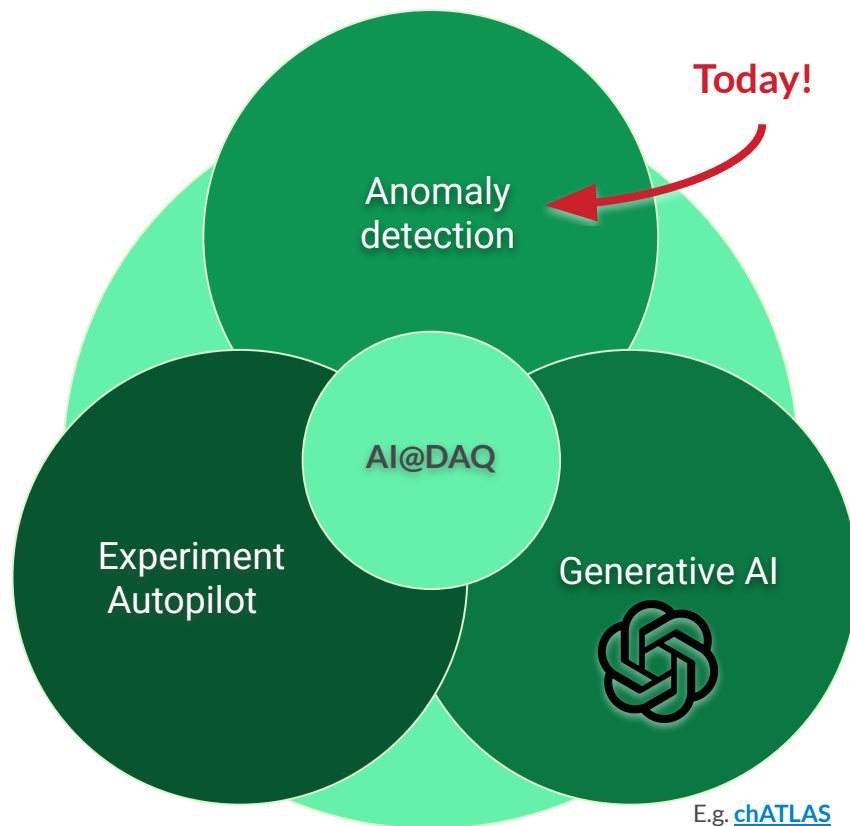


Disclaimers

- Tried to include all the work/plans from the LHC experiments
 - If your work is not here, let me know and I'm happy to include it!
- Anomaly Detection, even only on the DAQ has many topics ...
 - Main focus on **time-series AD** and most common methods

Introduction & Motivation

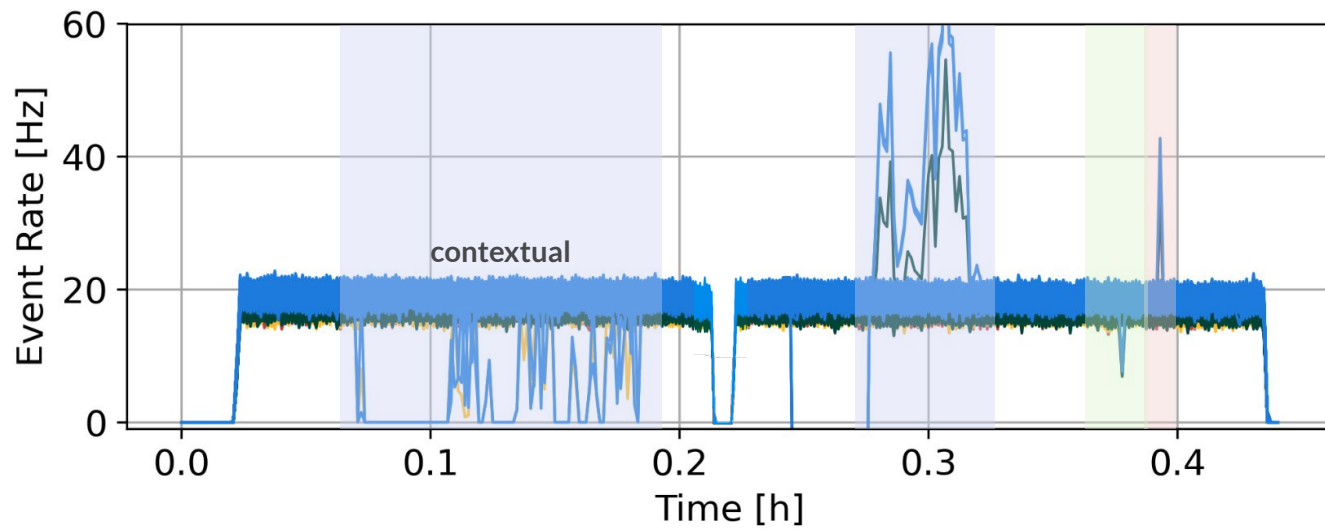
- **High-dimensional & complex data** in DAQ
 - Massive, high-rate data streams from multiple sub-detectors
- **Real-Time Needs:** Continuous monitoring for hardware, data quality, and safety
 - Classical AD methods (e.g., ARIMA¹, threshold rules) struggle with nonstationary and complex patterns
 - Need flexible, scalable, and robust approaches to capture subtle anomalies
- **Challenges:** Nonstationary conditions and subtle inter-signal correlations
- **Goal:** Detect anomalies early to minimize downtime and ensure reliable data taking



Project at CERN: **EP R&D WP9.1**

- Targeting real-time monitoring, with predictive analytics, increased efficiency and enhanced accuracy, using AI.

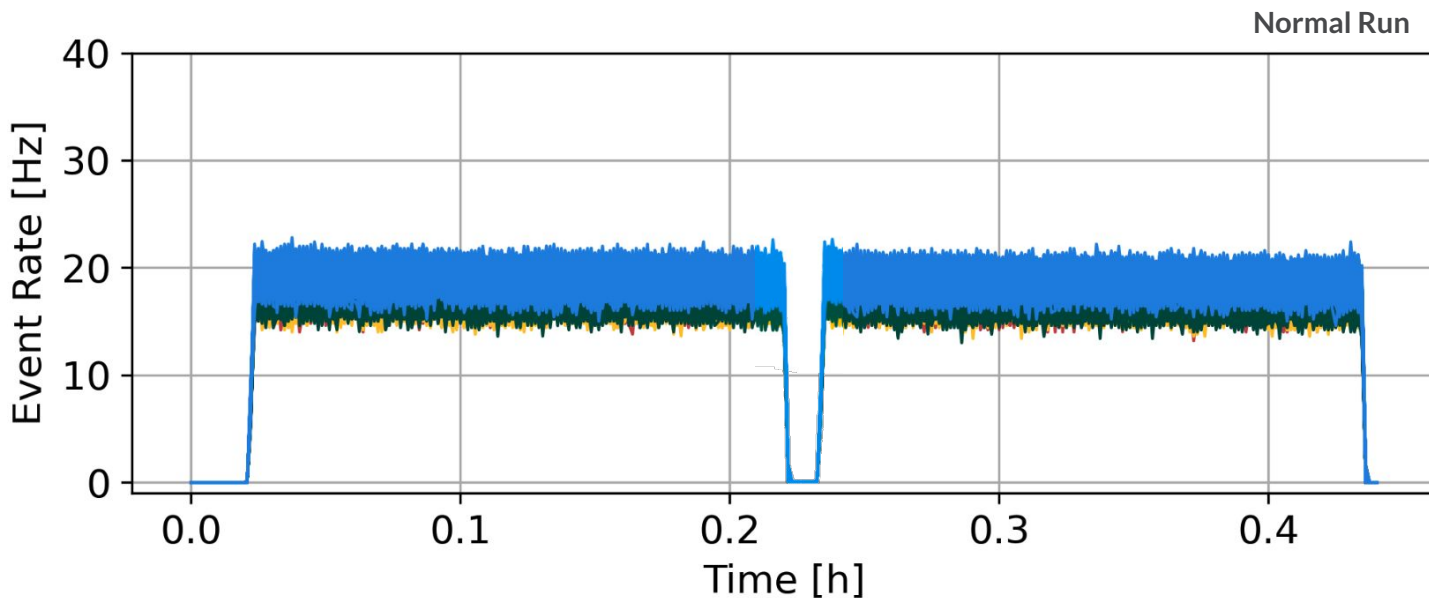
¹ AutoRegressive Integrated Moving Average



Textbook anomalies in Time Series

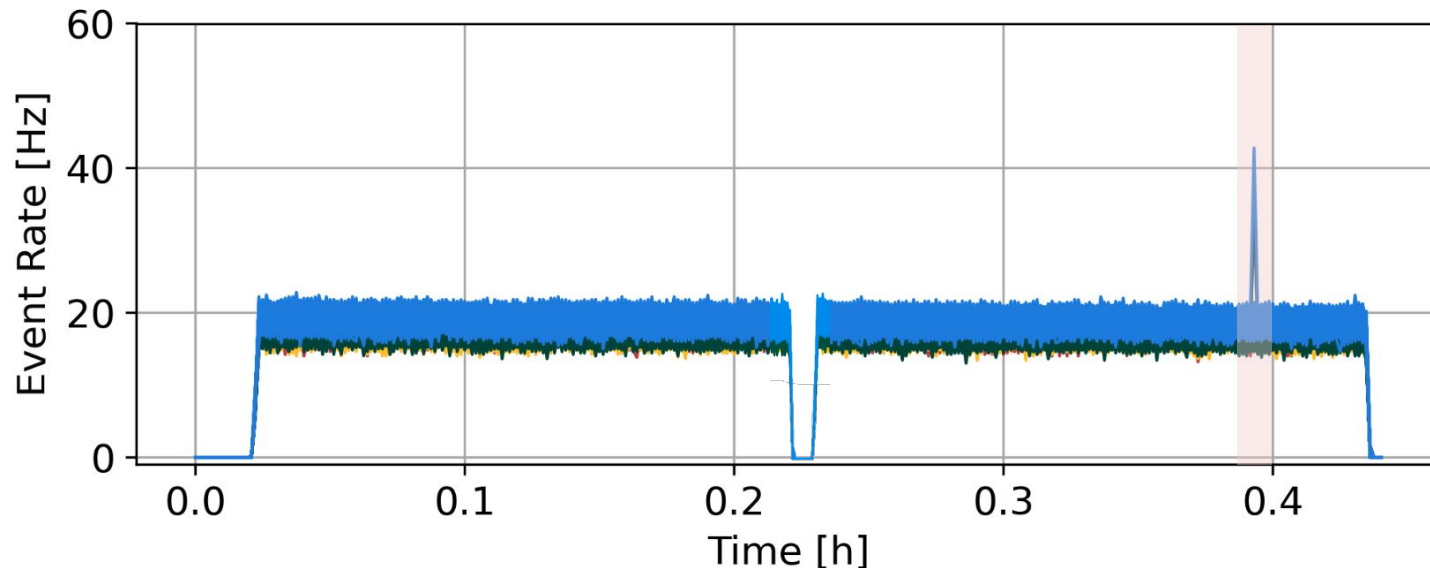
Time-Series *textbook* anomalies

- **Point Anomalies:** Single out-of-range data points
- **Contextual Anomalies:** Normal values in unusual context (e.g., time of day)
- **Persistent Anomalies:** Long-lasting deviations (e.g., sustained high temperature)
- **Collective Anomalies:** Group of points deviating as a pattern
- **Trend Anomalies:** Unexpected changes or breaks in trend direction
- **Seasonal/Periodicity Anomalies:** Deviations from expected cyclical patterns



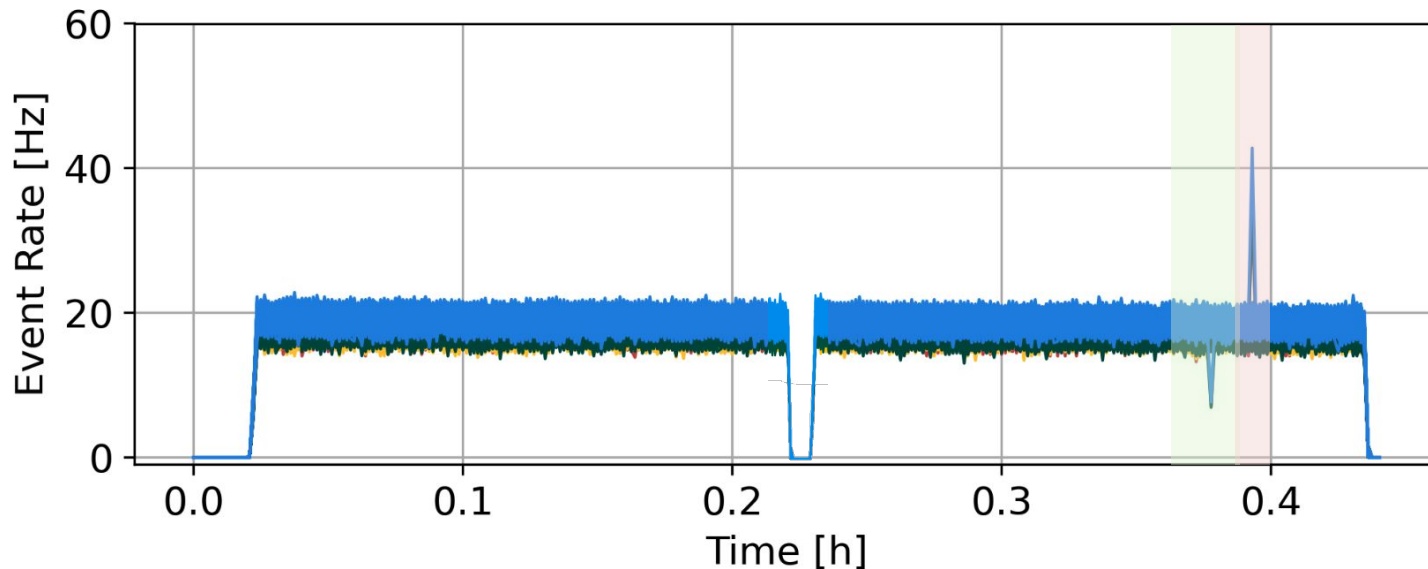
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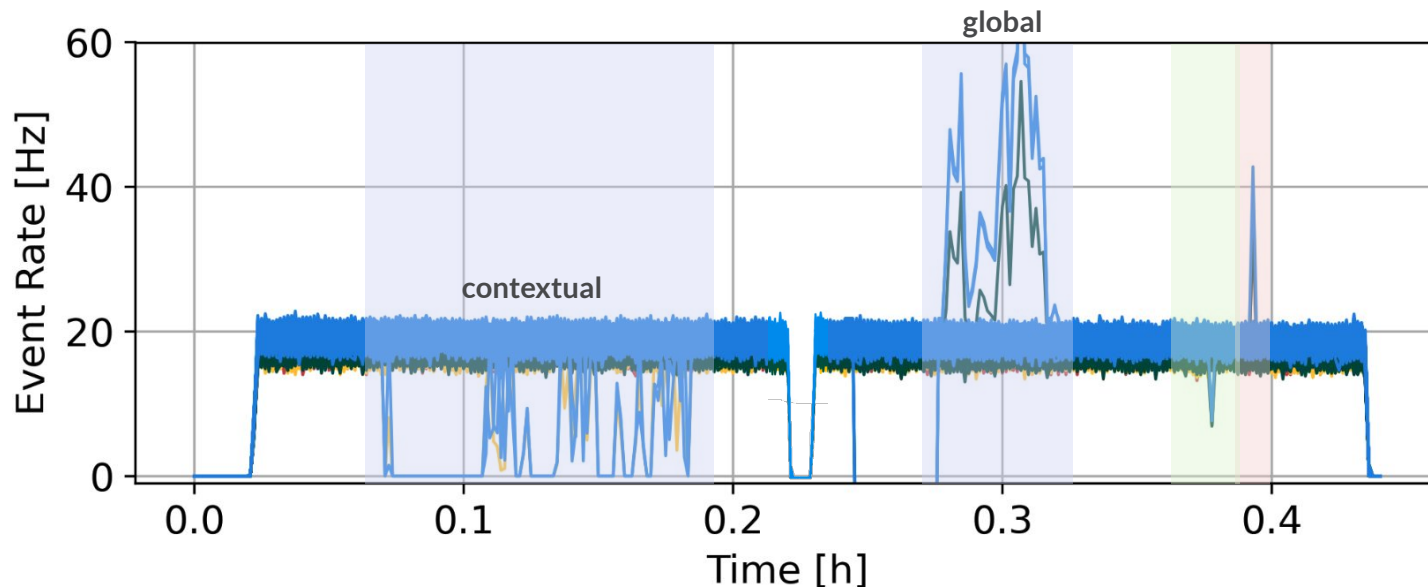
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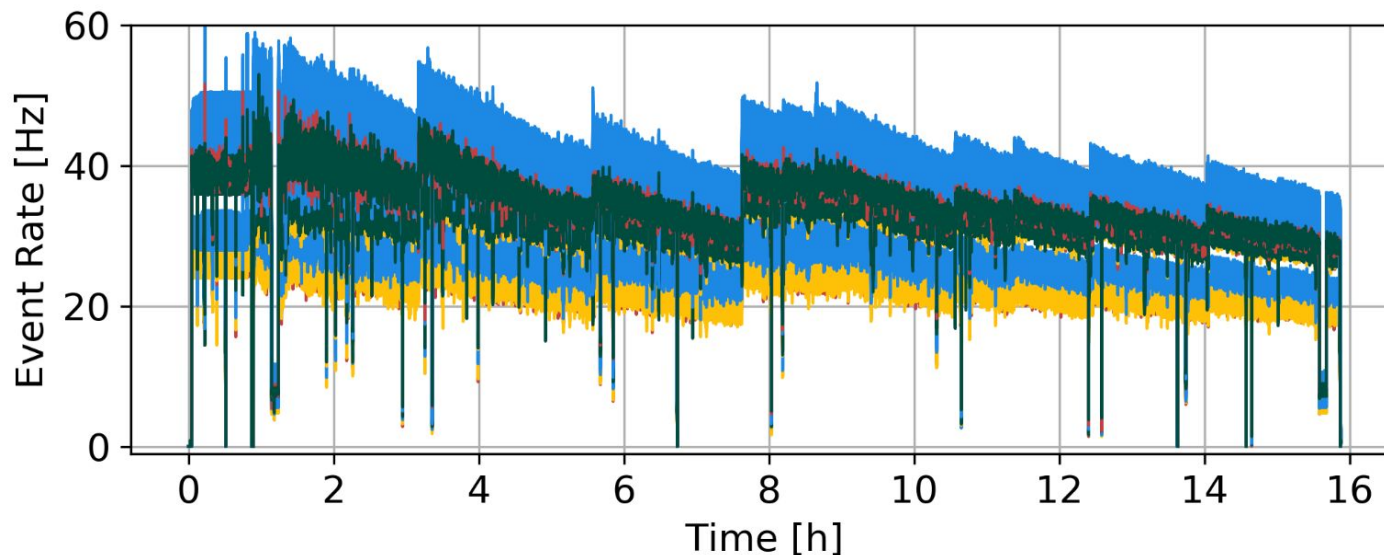


Time-Series *textbook* anomalies

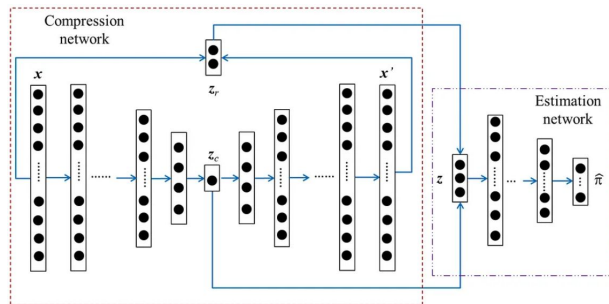
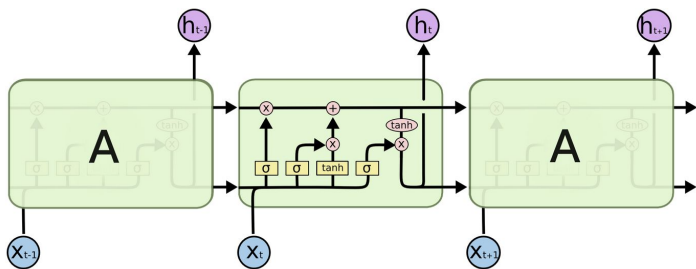
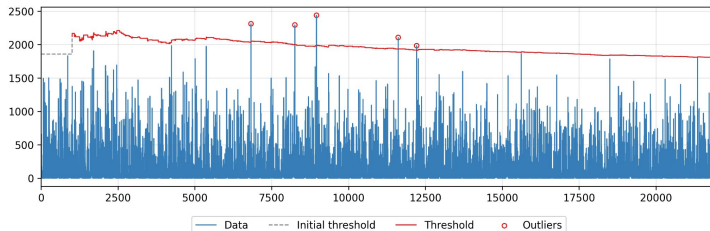
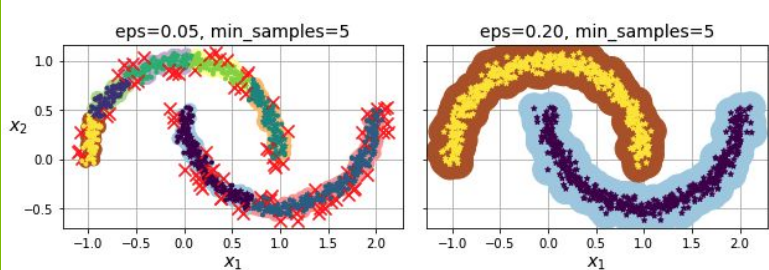
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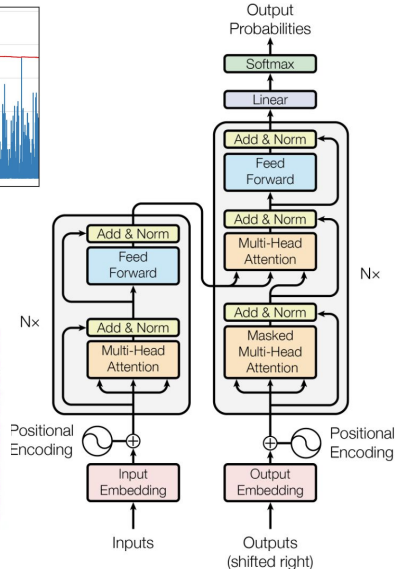
Example of a typical per-machine data event rate



- During physics data-taking session
- Individual channels (~2000) coloured by processor type
- Update interval is 5s



The whole process of DAGMM



ML Methods

Clustering, Deep Learning and evaluation metrics

Clustering Approaches

- Unsupervised identification of outliers in feature space
- Group similar patterns together; those that fall outside clusters are anomalies

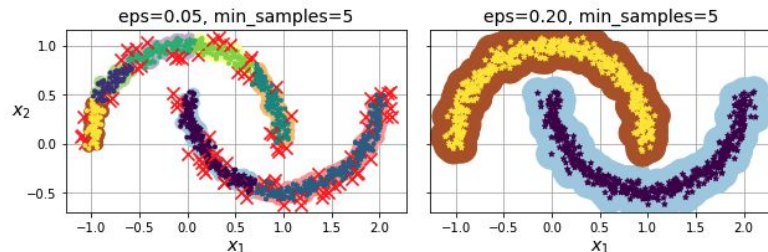
Common Techniques

- **Density-Based: DBSCAN**¹
 - Identify dense clusters, flag low-density points as outliers
- **Isolation Forest**
 - Randomly partitions feature space to isolate anomalies quickly
- **Clustering + Deep Embeddings**
 - Use deep neural networks (e.g., autoencoders) to embed data, then cluster in latent space for more robust anomaly detection

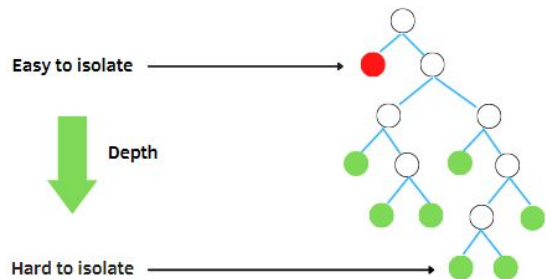
Advantages & Considerations

- Works well when anomalies are *globally* different from normal data
- Sensitive to parameter choices
- Can serve as an **initial stage** before deeper model-based analysis

DBSCAN [1]



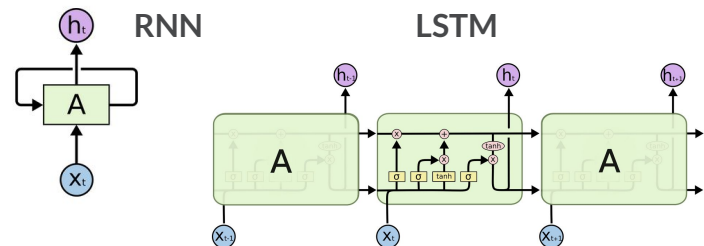
Isolation Forest [2]



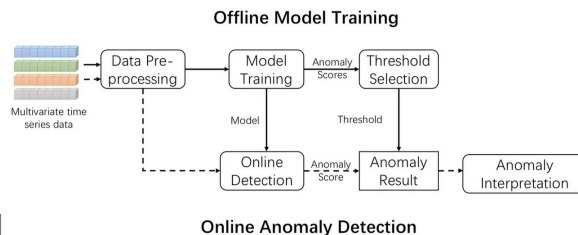
¹Density-Based Spatial Clustering of Applications with Noise

Deep Learning Approaches

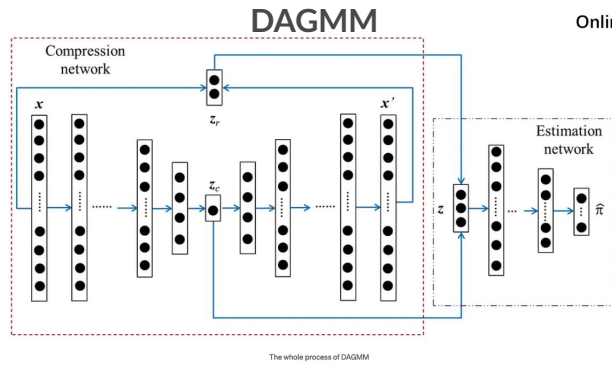
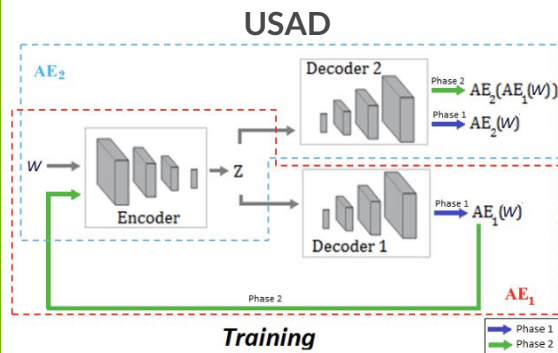
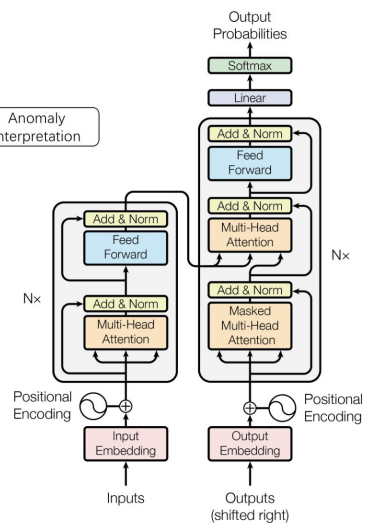
- **LSTM (Long-Short Term Memory)** [1]
 - Handles temporal dependencies, popular for sequential data
- **DAGMM (Deep Autoencoder Gaussian Mixture Model)** [2]
 - Learns latent representation + density estimation for anomaly scoring
- **USAD (UnSupervised Anomaly Detection)** [3]
 - Combines reconstruction and forecasting in a shared architecture
- **OmniAnomaly** [4]
 - VAE-based, learns probabilistic latent space for multivariate signals
- **Transformer-Based** (e.g., **TranAD**, **Informers**) [5] [6]
 - Uses attention for efficient long-term dependency modeling



OmniAnomaly



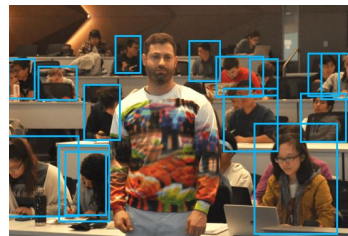
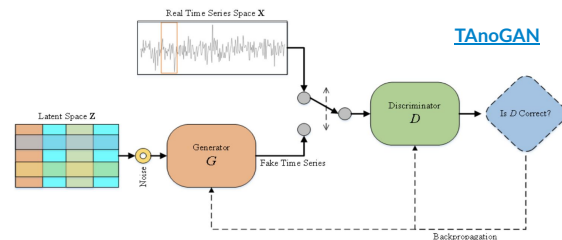
Transformers



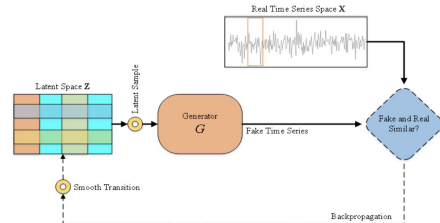
Generative Models & Reinforcement Learning

Generative Models

- **VAEs (Variational Autoencoders)**
 - Learn a latent distribution representing “normal” data
 - Reconstruction probability or likelihood can be used as anomaly score
- **GANs (Generative Adversarial Networks)**
 - Generator learns to create “normal-like” samples,
 - Discriminator learns to distinguish real from generated
 - Deviations from “normal” distribution flagged as anomalies



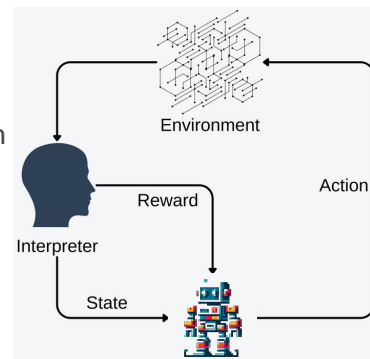
Adversarial clothing



Reinforcement Learning

- **Adaptive Thresholds**
 - RL agent learns a **policy** for setting dynamic anomaly thresholds to balance false positives/negatives
- **Active Exploration**
 - Potential to explore new data regimes and refine anomaly detection in changing conditions

Very interesting work done by the [CERN accelerator control](#) team.



Evaluation in Imbalanced Settings

- **Imbalanced Data**

- Few anomalies vs. huge normal dataset
- Careful validation with Precision, Recall, F1-score, **MCC**¹, etc

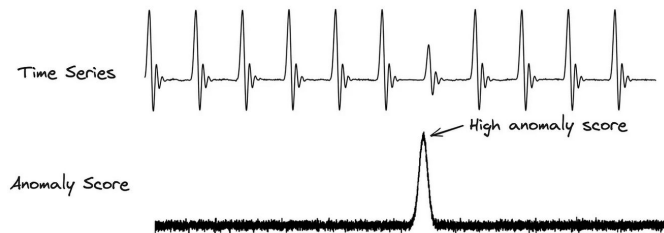
- **Threshold Methods**

Extreme Value Theory for adaptive thresholding

- **POT** (Peak Over Threshold): Process on the whole dataset to get one threshold
- **SPOT** (StreamingPOT): Initial POT but keeps updating the threshold

- **Model-Specific Evaluations**

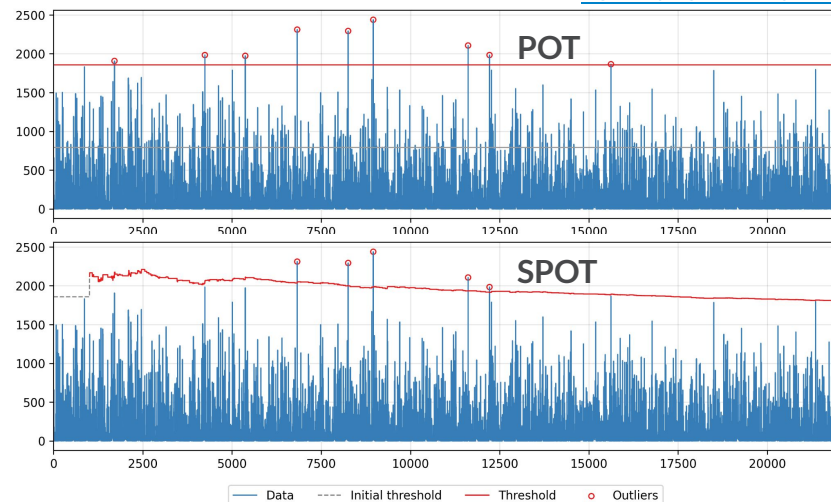
- **GANs**: Check real vs. generated distributions (e.g., detection error, reconstruction error)
- **Transformers**: Long-sequence performance, interpretability via attention weights
- **RL Approaches**: Reward-based metrics (trade-off between false positives & missed anomalies)



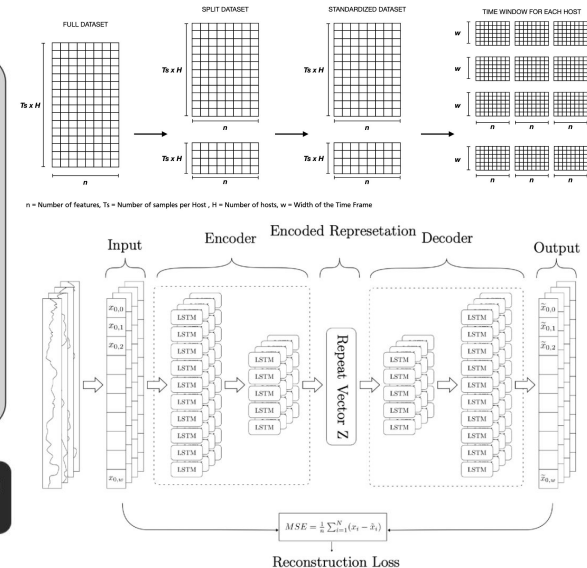
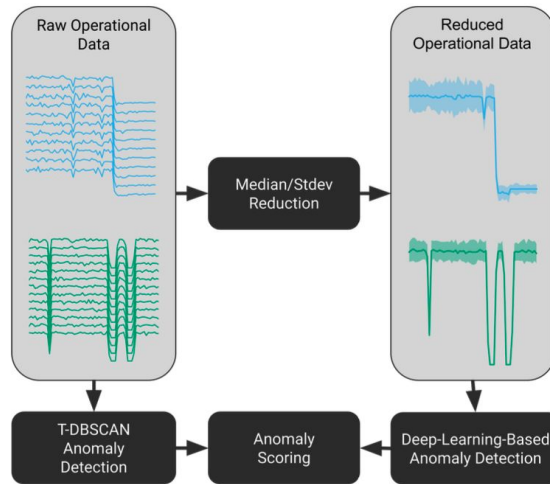
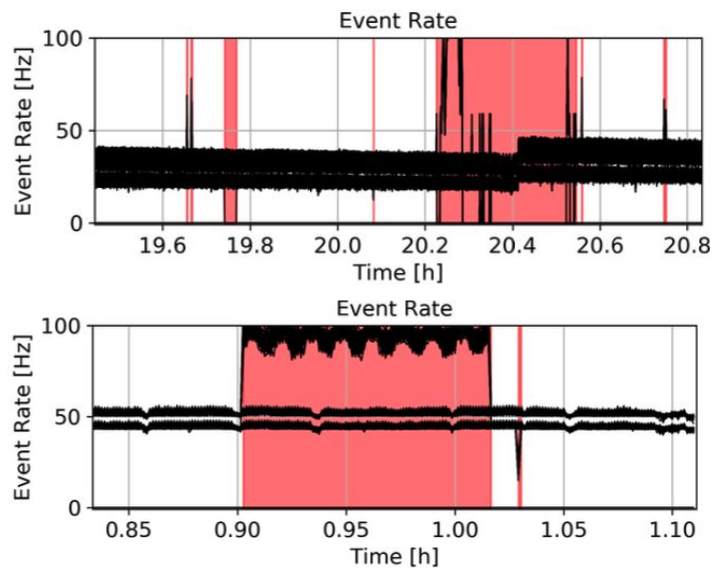
The output of an anomaly detector is the anomaly score.

[Image credit](#)

POT: Peak Over Threshold



¹ Matthews correlation coefficient



Ongoing work: status and plans

Anomaly Detection @ATLAS



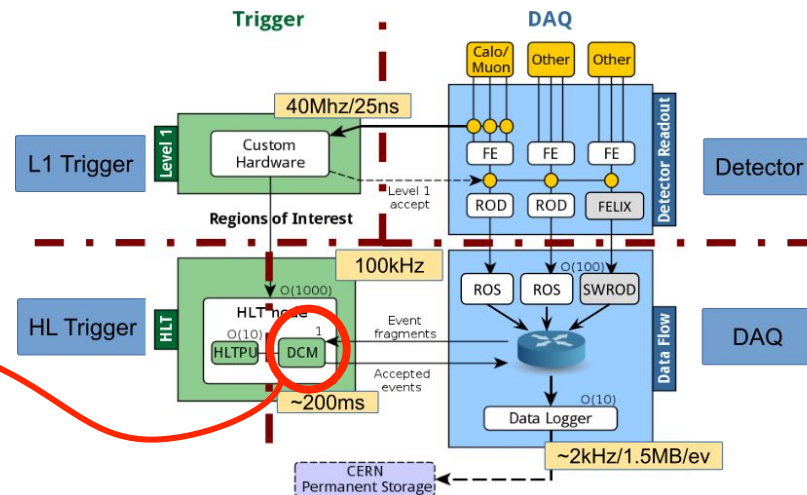
- Using **AD** in time series
 - Looking at a small part of the DAQ system, as a feasibility exercise
 - Initial work by K. Stehle et al. [DeepHYDRA](#)

DCM

retrieves the event fragments or the L1A and give this information to the **TPUs**.

Then builds the complete events that were selected.

- The HLT farm has highly variable configuration.
 - Large dimensionality ~2000 channels.
 - System is only locally homogeneous.
 - Operations continue after an anomaly.
 - Large variability in normal data.

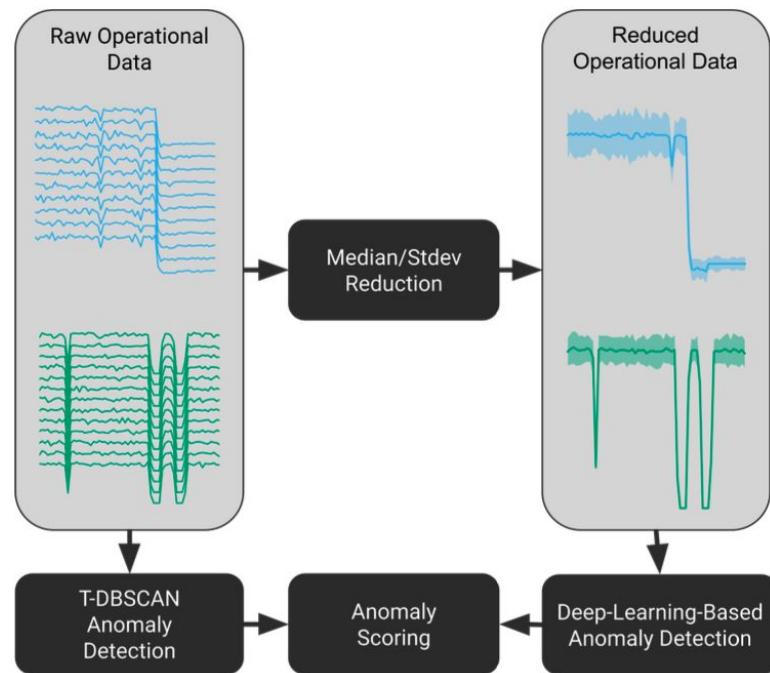


Other ATLAS work with time series (but outside the DAQ)
- [AE-based time series AD for ATLAS LAr calo. DQM](#)

AD@ATLAS: dataset and idea

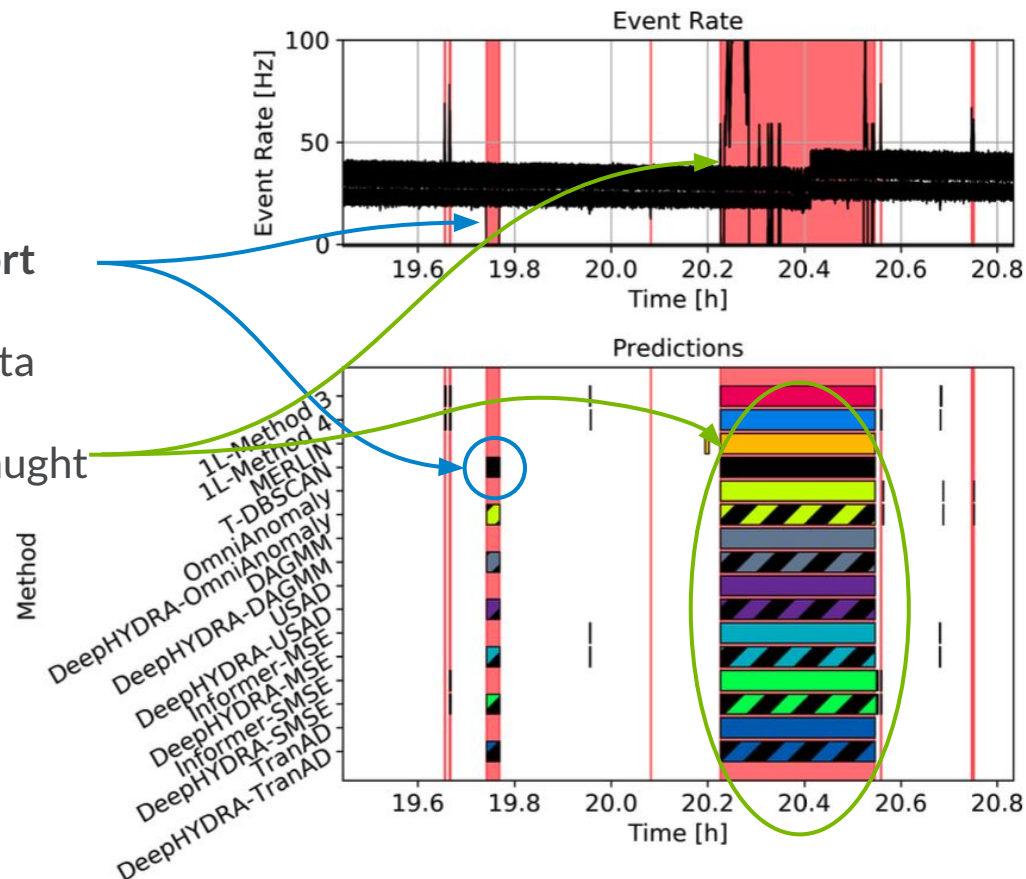
Kevin Stehle et al.

- Looking for anomalies using the **DCM** rate
- Full Run-2 and early Run-3 unlabelled data
 - Injected synthetic anomalies for testing
- **DeepHydra**: a hybrid approach
 - Clustering **T-DBSCAN**
 - Per-timestep detections
 - Each channel is converted as a data point
 - 1D problem with length = #nodes
 - Deep-learning-based
 - Long-range anomalies
 - Monitors the racks as a whole
 - Uses only **median** and **stdev**
 - Multiple models used ([see slide 12](#))



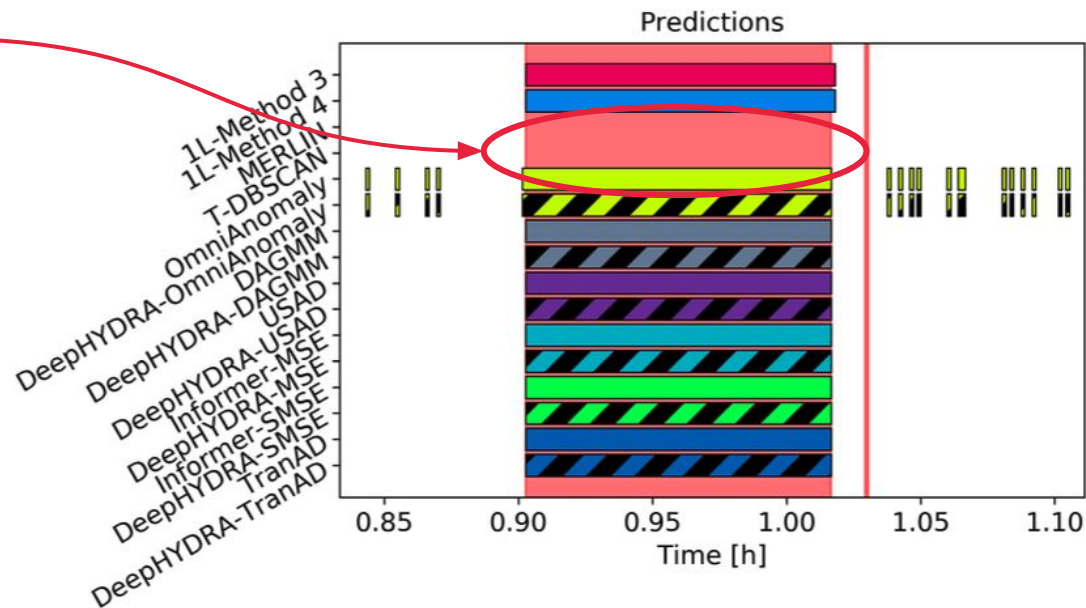
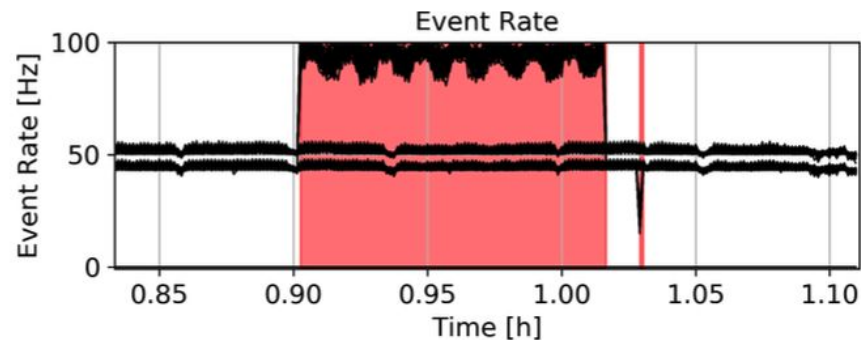
AD@ATLAS: Results

- The Hybrid approach proves to outperform individual attempts
- T-DBSCAN is able to identify a **short dropout** of a node having only a miniscule effect on the reduced data stream
- The long-term global anomaly is caught by all investigated models.



AD@ATLAS: Results - II

- On the other hand, **T-DBSCAN** can miss anomalies where all nodes in a subgroup show identical anomalous behaviour
- Here, the long-range deep learning catches the anomalies



AD@ATLAS: Plans

- Extend it to other parts of the **TDAQ**
 - Including Readout, data recording, etc
 - Can we detect issues in the **L1Trigger**? Anomalies in coverage due to problematic L1Muon/L1Calo triggers
- Can we provide an *ExpertAssistant*?
 - Similar to the ATLAS ShifterAssistant, but with no hardcoded rules by experts
 - Alerting the expert of a new or arising issue before a condition is met (if there is a condition...)
- Can we perform some operations/actions, based on the alerts produced by the AD system?

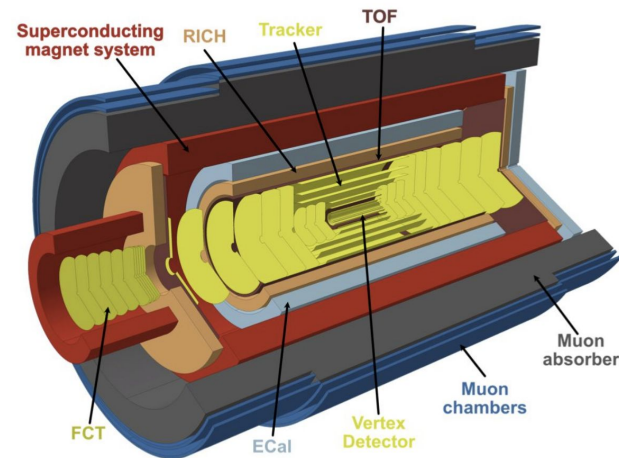
The screenshot displays the ELISA system's alert management interface. At the top, a navigation bar contains several tabs: 'Tabs per system', 'Run Control', 'Trigger', 'Shift/Laser', 'Murns', 'IO', 'Calorimeters', 'Lumi', 'DEBUIX', and 'Expert'. A 'Help' button is located on the right side of the navigation bar. Below the navigation bar, a status bar indicates 'Mark all alerts in this domain as read' and provides links for 'SUCCESS', 'INFORMATION', 'WARNING', 'ERROR', 'FATAL', and 'CRITICAL' alerts. The main content area is a table with columns for 'Read', 'Name & C', 'Message Q', and 'Action Q'. The table lists several alerts, including 'pixel in-error', 'partition-in-error', 'Initials ICT bad-messages', 'Initials ICT bad-messages', 'Initials ICT bad-messages', 'Initials ICT bad-messages', 'ICR-sign-check', and 'ICR-sign-check'. Each alert entry includes a 'Read' status, a 'Name & C' field, a 'Message Q' field, and an 'Action Q' field. A 'Notifications' panel on the right side of the screen shows a 'Login' button and a 'Logged in as' status. The bottom of the screen features a 'Actions' button and a 'Send it to the logbook' button.

- **Ultimate goal: Data Quality Monitoring**
 - Can we infer DQ issues with the data available in the DAQ (no histograms)?

ALICE status and plans



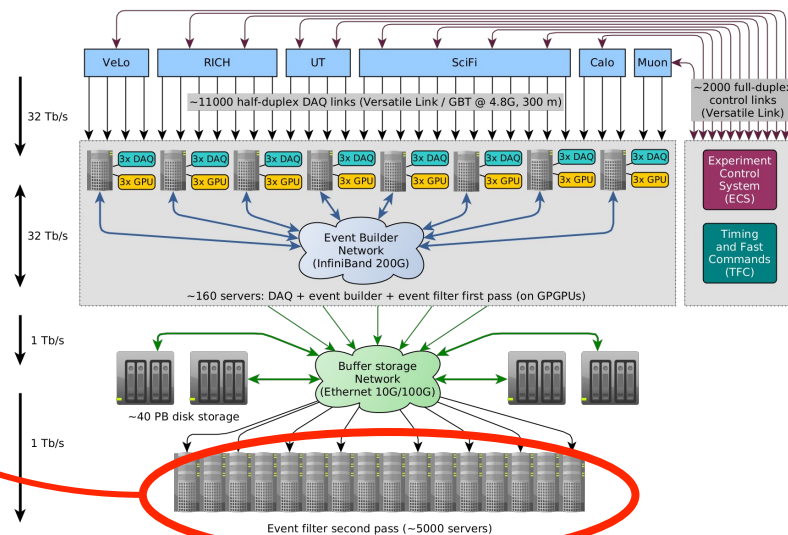
- Prototypes:
 - Inferring end-of-run reason from **application logs**
 - Memory usage analysis on readout nodes to **forecast out-of-memory** events
 - Anomaly detection in the **state machine transition time** of readout software
 - **Chatbot** to interface documentation / knowledge base, again in view of improving experiment operations
- Projects:
 - Automated Anomaly Detection in ALICE Inner Tracking System Data Using Machine Learning. [Student Report](#)
 - Defending ALICE with an anomaly-based intrusion detection system ([George Raduta master thesis](#))
- Plans:
 - Try using AI tools to ease experiment operations, which includes anomaly detection from logs and monitoring data



Anomaly Detection @LHCb

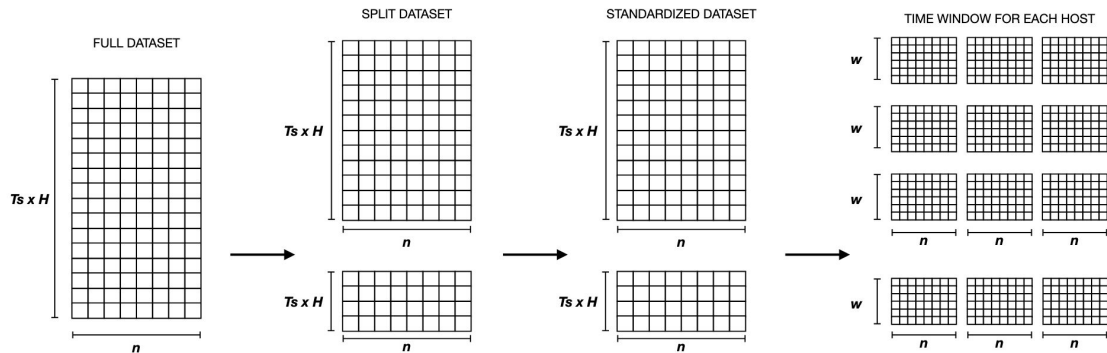


- Upgrade traditional monitoring systems relying on rule base alerts
 - In dynamic systems, **anomalies may appear normal** (e.g., CPU usage spikes from 0% to 100% during specific tasks)
- Use ML models to detect complex anomalies, in a **unsupervised** way, and provide insights to experts
- Looking at the **LHCb HLT2 Computing Infrastructure**
 - High Level Trigger 2 runs on CPUs on an heterogeneous computing infrastructure
 - ~5000 heterogeneous nodes with different specifications
 - All nodes perform the same computational task



- **4 months of system metrics.** Samples averaged over 20' intervals
 - CPU, memory, network, IOPS
 - Every action of every node is logged, can infer the states from there
 - Unlabelled dataset
- Pipeline for data-preprocessing
 - Cleaning, feature drop, normalization, validation and training split, and time windowing

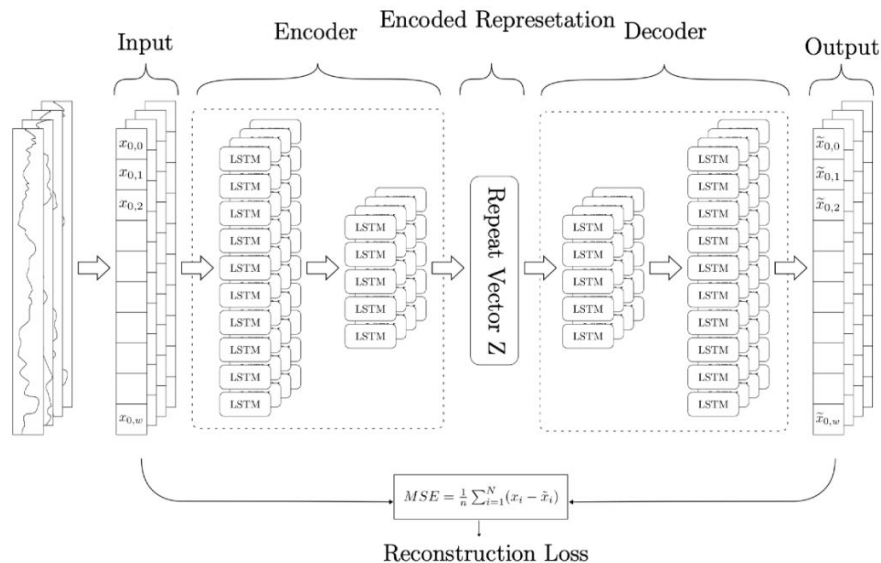
Time Windowing: Time windows of fixed size " $n \times w$ " are generated for both dataset. Each time window serves as a single input sample for the model.



n = Number of features, Ts = Number of samples per Host, H = Number of hosts, w = Width of the Time Frame

AD@LHCb: Model Architecture

- Use **autoencoders** to reconstruct input sequences and identify anomalies based on reconstruction errors
 - A large reconstruction error is associated with an anomaly
 - Reconstruction error is computed using **MSE**
 - Different Cell Types Used
 - LSTM
 - Gated Recurrent Unit (GRU)
 - Convolutional NN
 - Hyperparameter tuning
 - Number of layers and cells unit
 - Learning rate
 - Regularisation
- ~10-15% improvement

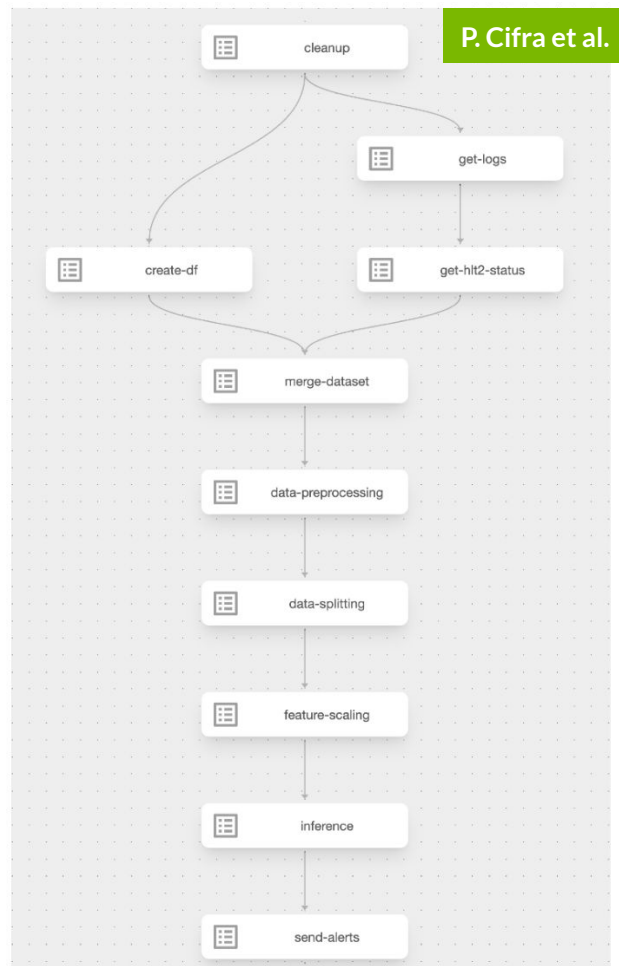


- A small benchmark dataset with **40k samples manually labelled** with known anomalies (50%)
 - Provide a ground truth for evaluating model's accuracy, precision, recall, and overall effectiveness
- **CNN Autoencoder** delivers the best overall results across metrics
 - GRU is the least complex, but training time was very similar

	Accuracy	Precision	Recall	F1-Score	AUC-ROC	PR-ROC
LSTM-AE	0.95	0.96	0.94	0.95	0.98	0.96
BI-LSTM-AE	0.95	0.95	0.94	0.95	0.97	0.96
GRU-AE	0.95	0.96	0.94	0.95	0.97	0.96
BI-GRU-AE	0.94	0.94	0.95	0.94	0.98	0.96
CNN-AE	0.96	0.96	0.97	0.96	0.98	0.97

AD@LHCb: Production System

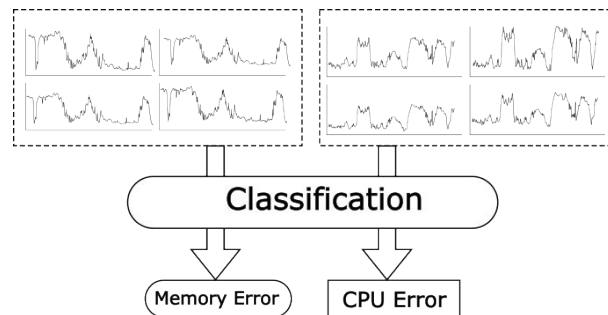
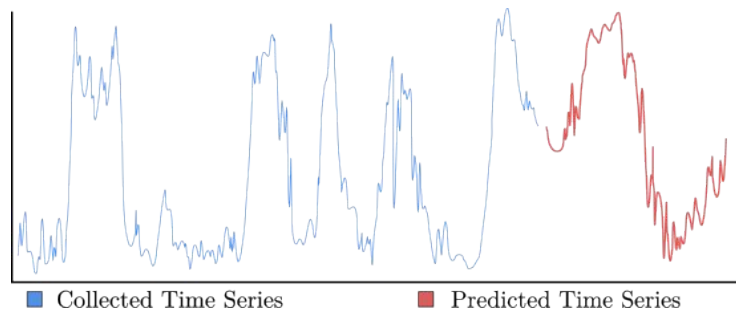
- Deployed in production using Kubeflow pipelines:
 - Kubeflow is an open-source platform for deploying, monitoring and managing ML models at scale
- Pipeline Overview
 - It is automatically executed every 4 hours
 - Replicate all the steps, from data collection and preparation to inference
- Alerting
 - If some nodes are flagged as anomalous, alerts are sent to the relevant experts



AD@LHCb: Plans

P. Cifra et al.

- Increase amount of data:
 - More months of data
 - Time frame overlap
 - More metrics
- Adapt model for different purposes:
 - Fault Prediction
 - Clustering similar anomalies
- Online training:
 - Update the model with “Real Time” data
- Alerting:
 - Better integration with the rest of the environment
 - Automatic actions in case of anomaly



Outlook

- Incorporation of **ML in the DAQ@LHC is ongoing**
 - First proofs of concept and preliminary results available
- All experiments looking at **AI** tools to **ease operations**, with real-time monitoring, predictive analytics, increasing efficiency and enhancing accuracy
 - We all have shortage of shifters in the control rooms → **more automation** should be really considered a priority at all the levels
- Having the same needs, and already doing similar or complementary work, provides a **great opportunity for joining efforts!**

Backup Slides



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ALICE

Anomaly-based Intrusion Detection System for ALICE deployments

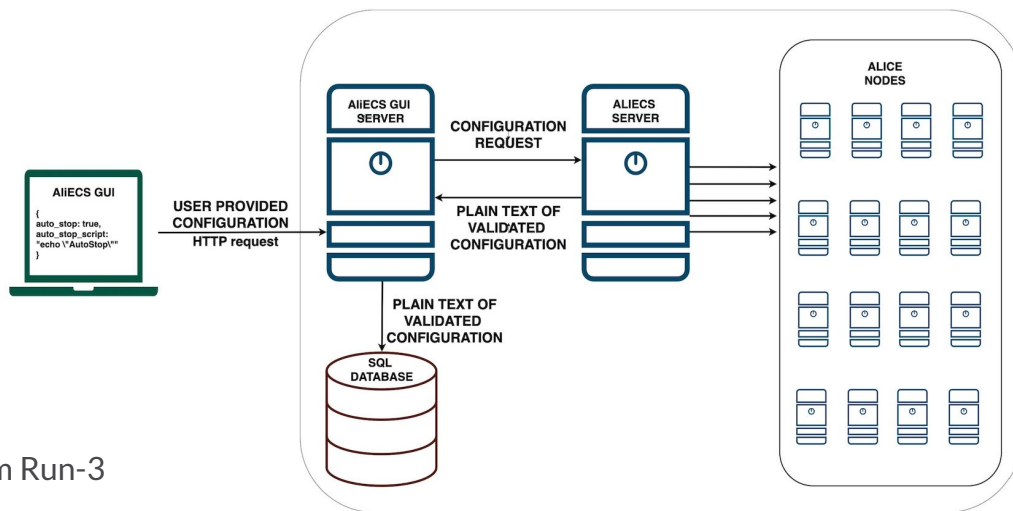
- LHC experiments require **precise** configurations; incorrect setups may cause hardware damage
- Conventional security systems focus on network traffic and login credentials, missing payload-based attacks
- The proposed system detects anomalous configurations before deployment

- Three Machine Learning Models have been tested:

- Isolation Forests (IF)
- **Autoencoder (AE)** - highest recall obtained
- Variational Autoencoder (VAE)

```
{
  "flag_1": false,
  "flag_2": "none",
  "flag_3": true,
  "abc_script": "AUTO_STOP_ENABLED=false;
  rm -rf /experiment/data; echo \"erasing data\""
}
```

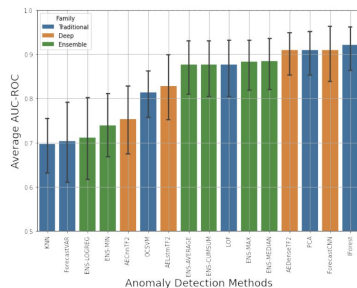
- Trained on thousands of historical configurations from Run-3
- **Threats effectively flagged:**
 - Unintentional misconfigurations by legitimate users
 - Malicious payload injections by malicious users



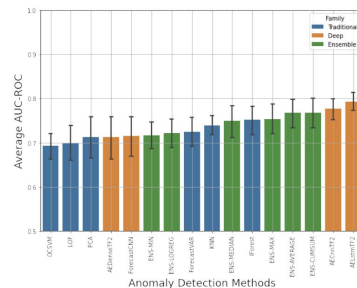
BONUS

Anomaly detection in the CERN cloud infrastructure

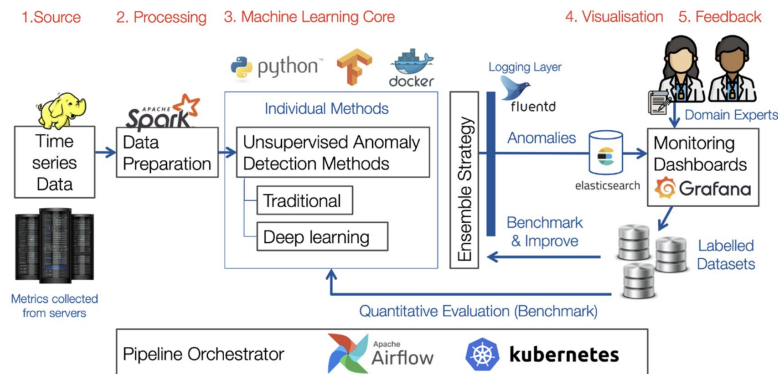
- Goal: reliably detect anomalies in the CERN cloud using time series monitoring system data (averaged over 12h)
- Many models tested
 - Traditional: Class SVM, LOF, Isolation Forest, PCA, KNN.
 - DeepLearning: AE FullyConnected, AE with CNN, AE with LSTM.



(a) "Batch" dataset



- Results reported in a JSON format, visualization and annotations in Grafana



Recurrent anomaly detection pipeline, implemented as Airflow DAG.

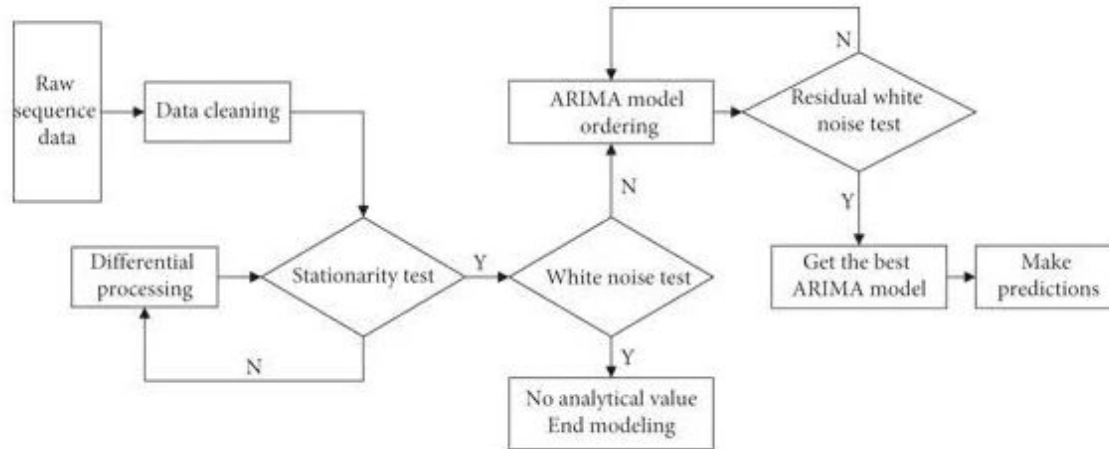
Legend: BashOperator, PythonOperator, queued, running, success, failed, up_for_retry, up_for_reschedule, upstream_failed, skipped, scheduled, no_status



ARIMA

AutoRegressive Integrated Moving Average

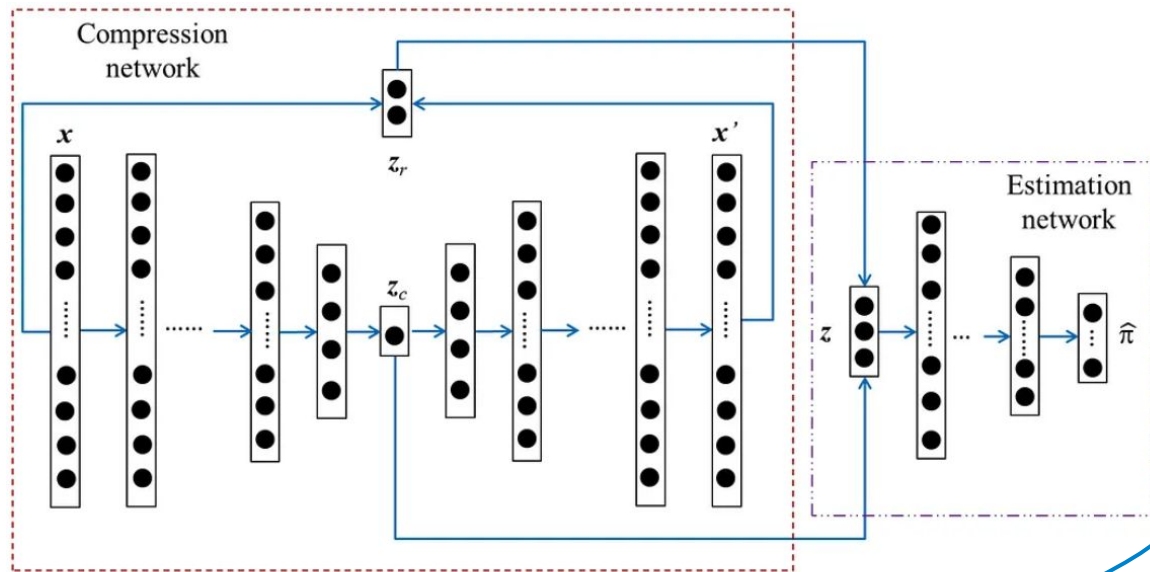
ARIMA is a classical statistical method for time series forecasting. It combines autoregressive (AR) models, differencing (to make the data stationary), and moving average (MA) models.



Can be extended to SARIMA, considering seasonal effects.

DeepLearning: DAGMM

The Deep Autoencoding Gaussian Mixture Model operates in two distinct steps: the **Compression Network** and the **Estimation Network**.



The whole process of DAGMM

Compression Network

Uses an **AutoEncoder** to generate a latent vector (Z_c) and compute the reconstruction error. These two outputs are then concatenated to form a new vector 'z' (input for the Estimation Network).

Estimation Network

Performs density estimation using a **Gaussian Mixture Model (GMM)** framework.

DAGMM objective function:

- 1- The reconstruction error from the AutoEncoder in the Compression Network.
- 2- The energy function, represents the likelihood of observing each data point at a specific Gaussian distribution.
- 3- Addresses the singularity problem in GMM.

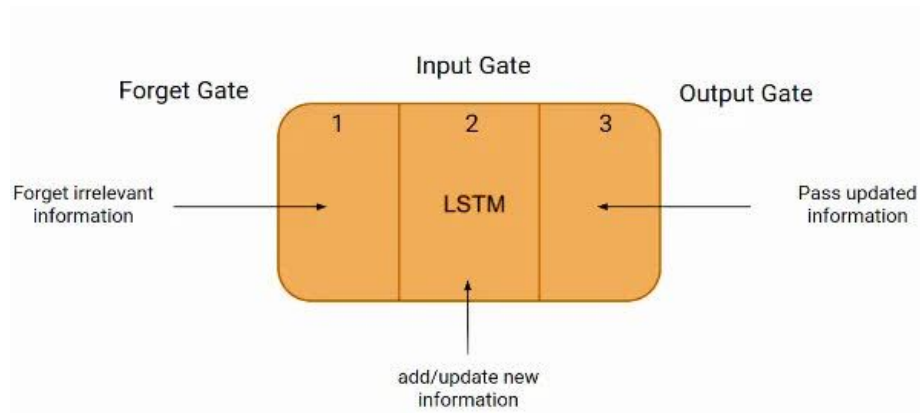
$$J(\theta_e, \theta_d, \theta_m) = \frac{1}{N} \sum_{i=1}^N L(\mathbf{x}_i, \mathbf{x}'_i) + \frac{\lambda_1}{N} \sum_{i=1}^N E(\mathbf{z}_i) + \lambda_2 P(\hat{\Sigma}).$$

LSTM

Long Short-Term Memory

A type of recurrent neural network (RNN) capable of learning order dependence in sequence prediction problems.

- **Forget Gate**
Decides the information that should be discarded from the cell state.
- **Input Gate**
Updates the cell state by deciding which values to update.
- **Output Gate**
Decides what the next hidden state should be, which contains information on previous inputs.



The combination of these gates allows LSTMs to mitigate the vanishing gradient problem typical in traditional RNNs, where gradients decay exponentially over time or sequences, making it hard to learn long-range dependencies.

Many other methods

- **Prophet**

- Developed by Facebook, is a tool for forecasting time series data that can handle missing data and outliers and provide reliable uncertainty intervals.

- **WaveNet**

- Developed by DeepMind, is a deep generative model designed for audio generation but can be used for time series forecasting.

- **DeepAR**

- Developed by Amazon, is a probabilistic forecasting algorithm for time series prediction. It uses recurrent neural networks (RNNs), to model complex time series datasets effectively.

- **Seq2Seq**

- are a type of NN architecture developed for tasks where both input and output are sequences.

- **GRU (Gated Recurrent Unit)**

- is a type of RNN that is used primarily for sequential data processing, similar to the LSTMs but with a simpler architectural design.

- **XGBoost**

- is a gradient boosting method that can be used for time series prediction by transforming the problem into a supervised learning one, treating previous time steps as features.

Clustering: DBSCAN

Density-Based Spatial Clustering of Applications with Noise. Groups points in a dataset that are closely packed together, based on their distance to other points.

Characteristics

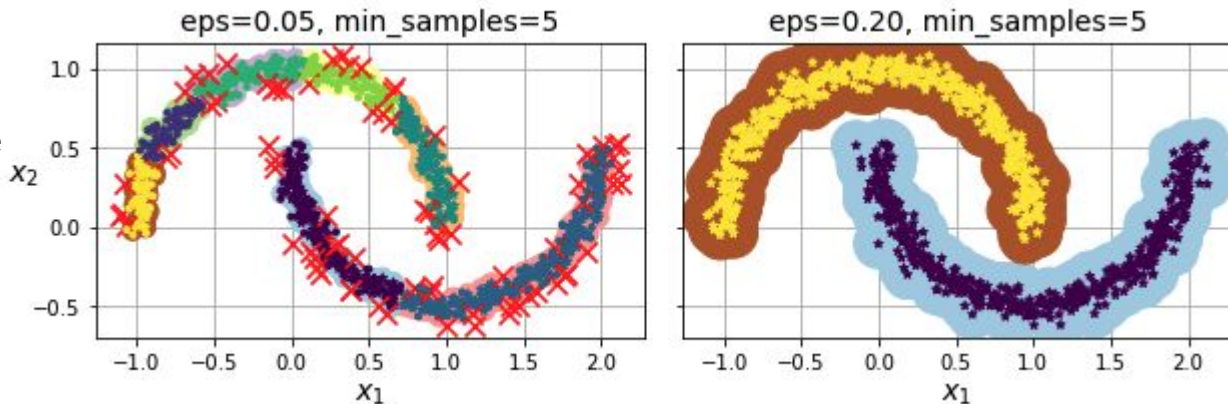
- + capable of identifying clusters of any shape.
- + robust to noise (good for outlier detection).
- + is a parameter-free clustering algorithm (no need for # clusters).
- + efficient on large datasets with a time complexity of $O(n \log n)$.
- its performance can be sensitive to the choice of hyperparameters.
- may perform poorly on datasets with clusters of significantly different densities.
- performance can be limited on high-dimensional data due to the "curse of dimensionality".

```
from sklearn.cluster import DBSCAN
from sklearn.datasets import make_moons

X, y = make_moons(n_samples=1000, noise=0.05, random_state=42)
dbscan = DBSCAN(eps=0.05, min_samples=5)
dbscan.fit(X)
```

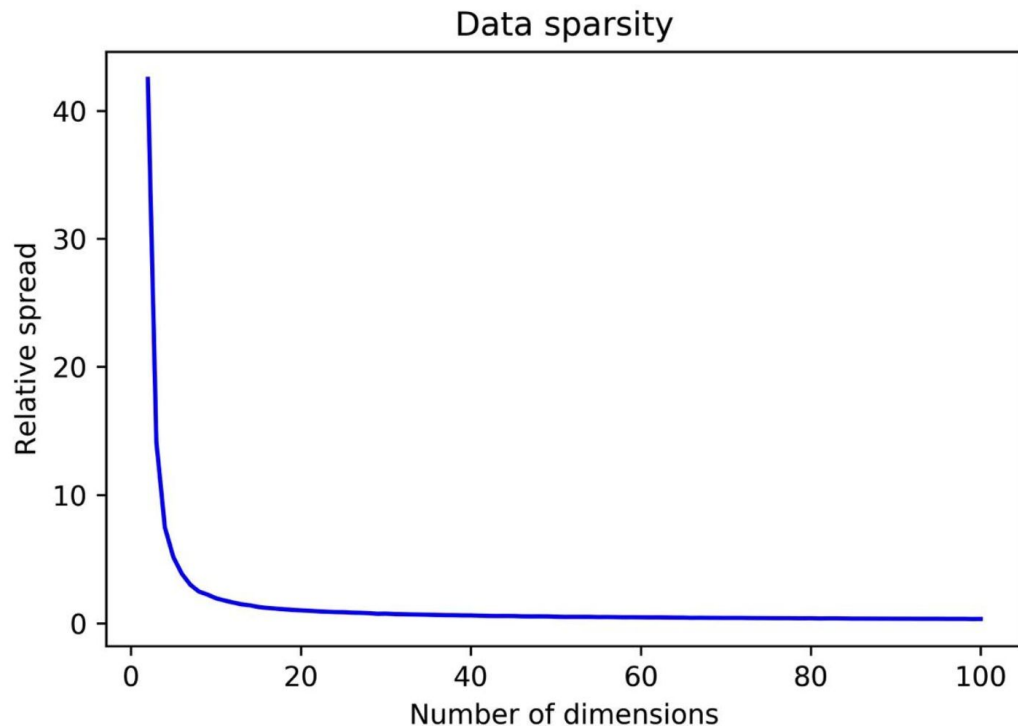
eps: maximum distance between two points for them to be considered neighbors. Points that are within eps distance of each other are considered part of the same cluster.

min_samples: minimum number of points required for a point to be considered a core point. Points that have fewer than min_samples neighbors are labeled as noise.



[Hands-on Machine Learning with Scikit-Learn, Keras and TensorFlow \(3rd edition\)](#)

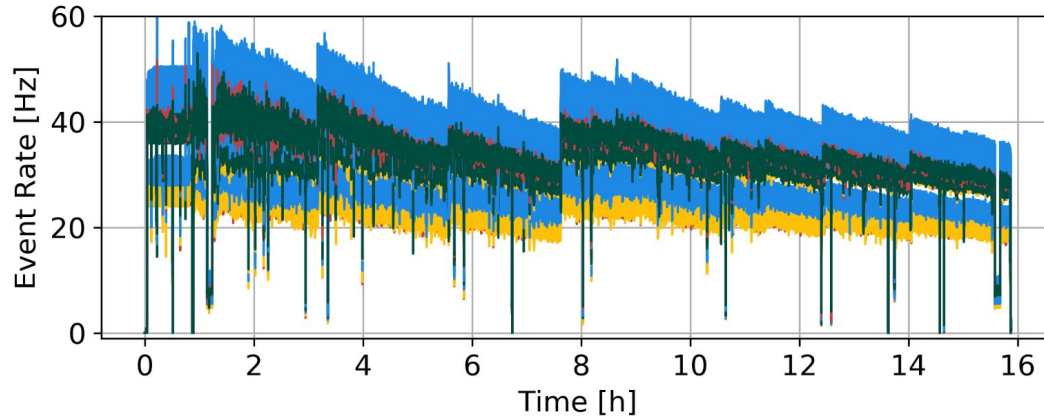
The curse of dimensionality



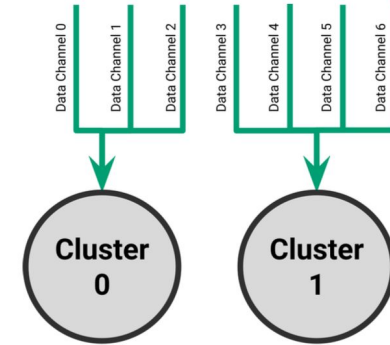
For a randomly chosen point, the relative difference between the furthest-neighbor distance r_{max} and nearest-neighbor distance r_{min} goes to **zero** as the dimensionality d grows:

$$\lim_{d \rightarrow \infty} \left(\frac{r_{max} - r_{min}}{r_{min}} \right) = 0$$

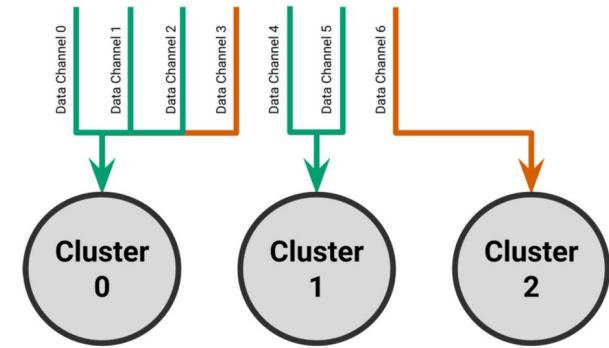
ATLAS: Data and Clustering example



- Example of a typical per-machine data event rate in a physics data-taking session.
- Individual channels coloured by processor type.
- Update interval is 5s.



Example 1: No Anomalies in the Input Data.



Example 2: Anomalies in Data Channels 3 and 6.

AD@ATLAS: Results - Metrics

DeepHYDRA is our hybrid approach

- Always outperforms the individual version
- The less computationally intensive models **DAGMM** and **USAD** outperform the transformer-based versions in the hybrid scenario

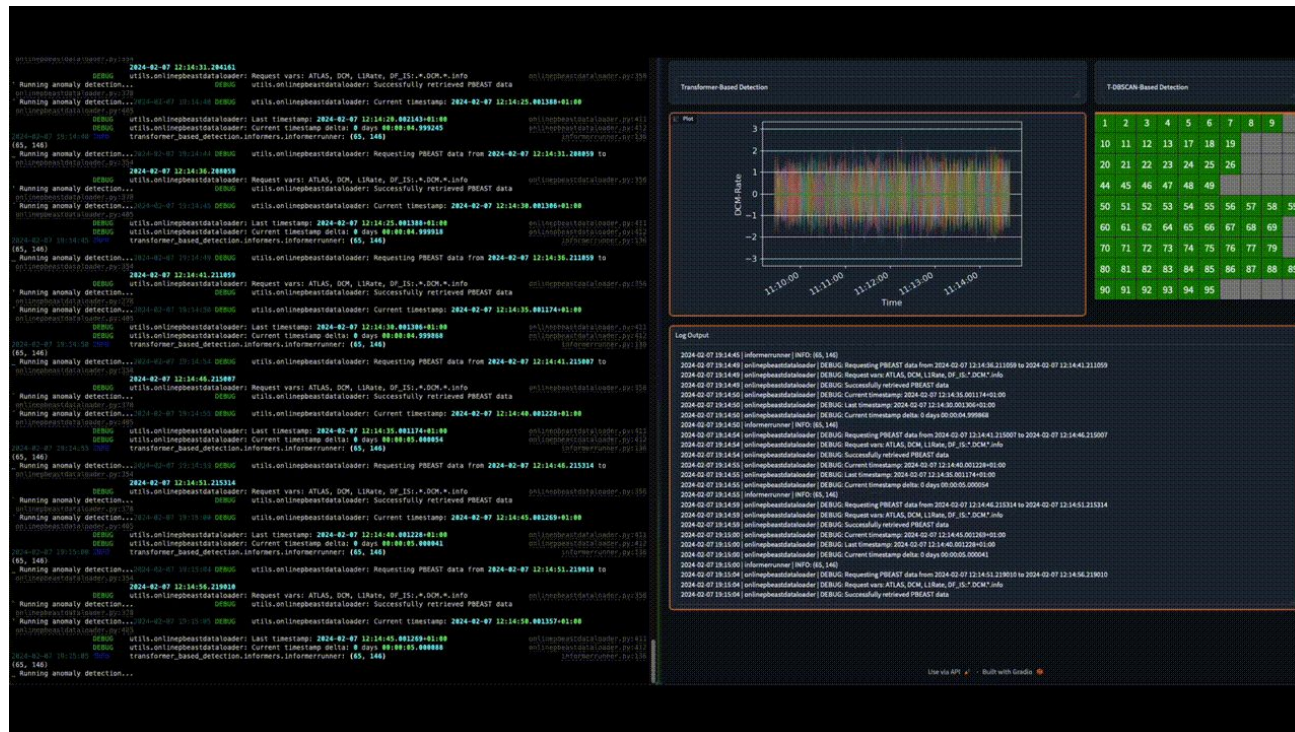
Model	MCC (mean/std)	MCC (best)	F1 (best)	AUC-ROC (best)
1L-Method 3 (> 45 %)	-	82 %	84 %	92 %
1L-Method 4 (> 47.5 %)	-	84 %	86 %	93 %
MERLIN-S (> 72.5 %)	-	72 %	75 %	86 %
MERLIN-P (> 72.5 %)	-	72 %	75 %	86 %
T-DBSCAN	-	79 %	80 %	83 %
OmniAnomaly	70.7 % ± 3.9 %	77 %	79 %	91 %
DeepHYDRA-OmniAnomaly	78.7 % ± 2.6 %	82 %	84%	95 %
DAGMM	78.8 % ± 2.1 %	80 %	81 %	85 %
DeepHYDRA-DAGMM	89.7 % ± 2.6 %	92 %	92 %	94 %
USAD	75.3 % ± 9 %	84 %	86 %	90 %
DeepHYDRA-USAD	84.9 % ± 5.2 %	92 %	93 %	96 %
TranAD	78.9 % ± 0.9 %	80 %	83 %	89 %
DeepHYDRA-TranAD	86.5 % ± 1 %	88 %	89%	95 %
Informer-MSE	81.8% ± 1.1 %	84 %	86 %	92 %
DeepHYDRA-MSE	88.1 % ± 0.9 %	89 %	91 %	97 %
Informer-SMSE	83.7 % ± 0.5 %	85 %	86 %	93 %
DeepHYDRA-SMSE	90.2 % ± 0.7 %	91 %	92 %	97 %

(c) Unreduced HLT Test Set

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP} + \text{FP}) \cdot (\text{TP} + \text{FN}) \cdot (\text{TN} + \text{FP}) \cdot (\text{TN} + \text{FN})}}$$

AD@ATLAS: Online detection

- Current implementation running on the Lab, looking at P1 information (this is for testing, large delay between the lab and the Control Room)



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